ChatGPT

Open-Source Windows GUI Control for AI Agents

Executive summary

The open-source Windows GUI automation landscape is real, but it is fragmented into three distinct layers. The first layer is **accessibility-first automation**, built on Microsoft UI Automation and older accessibility stacks; this is where the most mature Windows-native tools live, including pywinauto, FlaUI, WinAppDriver, and Python UIAutomation wrappers. The second layer is **image/input-first automation**, such as PyAutoGUI, SikuliX, and RobotJS, which works even when applications expose little or no accessibility metadata, but is naturally more brittle under DPI changes, theming, localization, and layout drift. The third layer is the new **agent/MCP layer**, where projects such as Windows-MCP and MCPControl expose desktop control to LLMs through MCP or similar protocols; these are promising, but today they mostly wrap lower-level primitives rather than providing the equivalent of a Playwright-quality native Windows driver. [1]

The most important technical conclusion is that a serious Windows GUI control stack should be **hybrid by design**. UI Automation gives the best selectors and the safest semantics when controls expose a good automation tree, including properties, control patterns, and events. But UIA coverage is uneven across legacy Win32 apps, custom-drawn controls, remote/virtualized surfaces, browser edge cases, icon-only toolbars, and GPU-heavy or game-like UIs. A perception fallback using OCR, layout analysis, and lightweight GUI element detection is therefore not optional if the goal is broad agent coverage. Recent agent-oriented work such as OmniParser, OSWorld, and Windows Agent Arena reinforces that modern “computer-use” systems need both a structured action interface and screen understanding. [2]

Windows security and privilege boundaries are the main architectural constraint. UIPI prevents lower-integrity processes from sending many messages or injected input into higher-integrity targets, and Microsoft’s documented UIAccess path is narrowly scoped to trusted assistive technologies, with signing, secure installation location, and manifest requirements. In practice, that means a production-grade open-source controller should default to **non-elevated automation**, treat elevation as an explicit opt-in mode, and keep privileged helpers separate from the main daemon. [3]

The best open-source-friendly product direction is therefore a **two-layer design**: a deterministic Windows controller service modeled more like WebDriver or Playwright, and an optional MCP shim that exposes agent-friendly tools on top of it. MCP is well-suited to tool interoperability, while WebDriver remains the clearest precedent for a language-neutral remote control interface. A native Windows controller should apply the same separation of concerns to desktop automation. [4]

Open-source project inventory

The table below focuses on projects directly relevant to programmatic control of Windows desktop GUIs, plus a few adjacent harness/perception projects that matter for AI-agent use. “Accessibility/UIA/MSAA/IA2” means the project directly consumes Windows accessibility APIs or clearly exposes selector semantics based

on those APIs. “Image-based” means template matching, screenshot parsing, or vision-first operation. Links are in the **Sources** column.

Project	Category	License	Primary language	Element-level control	Image-based automation	Accessibility / UIA / MSAA / IA2	Low-level input
pywinauto	Windows automation library	BSD-3-Clause	Python	Yes	No	Yes; Win32 and UIA backends	Yes; keyboard/ mouse modules
FlaUI	Windows automation library	MIT	C#	Yes	No	Yes; wrapper over native UI Automation	Partial; usually through UIA-driven actions
WinAppDriver	Remote automation driver	MIT	C++ / C# service with multi-language clients	Yes	No	Not explicitly stated in surfaced repo text, but exposes Selenium-like Windows element automation	Limited; action layer is WebDriver-style
Python UIAutomation for Windows	Windows automation library	Apache-2.0 inferred from package ecosystem; exact SPDX not re-opened in this pass	Python / C++ extension	Yes	No	Yes; Windows UI Automation by design	Partial
AutoHotkey	Scripting / macro automation	GPL-2.0	C++ + AHK script	Limited semantic control	No	No	Yes; strong hotkey/ input automation

Project	Category	License	Primary language	Element-level control	Image-based automation	Accessibility / UIA / MSAA / IA2	Low-level input
PyAutoGUI	Image/ input automation	BSD-3-Clause	Python	No	Yes; basic image recognition	No	Yes
SikuliX / OculiX continuation	Vision-first desktop automation	License not verified in surfaced sources here	Java + Jython/ Ruby scripting	No	Yes; OpenCV-based	No	Yes
RobotJS	Desktop input/ screen library	MIT	Node.js / native addon	No	Partial; screen read only	No	Yes
TestStack.White	Legacy Windows automation library	Mixed / unclear on repo page	C#	Yes	No	Yes; based on UI Automation	Partial
Windows-MCP	MCP desktop control server	MIT	Python	Yes; via snapshot/ state tools	Yes; screenshot-first and optional vision	Yes; UI state, browser DOM mode, Firefox IA2 fallback	Yes
MCPControl	MCP desktop control server	MIT	TypeScript / Node.js	Limited; focus is operations over mouse, keyboard, windows	Partial; screen capture/ analysis	Not clearly exposed in surfaced docs	Yes
MCP Windows Desktop Automation	MCP wrapper over AutoIt	MIT	TypeScript	Yes; via AutoIt control functions	Partial; screenshots	No	Yes

Project	Category	License	Primary language	Element-level control	Image-based automation	Accessibility / UIA / MSAA / IA2	Low-level input
Windows Agent Arena	AI harness / benchmark	License not verified in surfaced sources here	Python	N/A as a library; benchmark environment	Multimodal agent setting	N/A	N/A
OSWorld	AI harness / benchmark	License not verified in surfaced sources here	Python	N/A as a library; benchmark environment	Multimodal agent setting	N/A	N/A
OmniParser	Perception module for GUI agents	CC-BY-4.0	Python	Detects GUI elements, but does not control them	Yes; pure vision screen parsing	No; vision-first	No

What the table shows is that **maturity still clusters around accessibility-first libraries**, not around MCP-native or AI-native systems. pywinauto, FlaUI, WinAppDriver, and older White-style stacks exist because UI Automation still offers the most reliable semantic grounding on Windows when providers are implemented well. By contrast, the MCP projects are younger and mostly concentrate on tool exposure, orchestration, and screenshot/state packaging rather than on a deep native element model. [20]

The adjacent harness/perception projects matter because they show where the field is going. Windows Agent Arena and OSWorld provide evaluation infrastructure for multimodal desktop agents, while OmniParser directly targets screenshot-to-element grounding. Together they imply that a future-proof Windows controller should not stop at UIA wrappers; it should produce structured state, screenshots, and fallback perception hooks suitable for agent loops and benchmarking. [21]

Windows control interfaces

Windows exposes several overlapping control surfaces. The safest way to think about them is as a stack: **semantic accessibility APIs** at the top, **HWND/window/message APIs** in the middle, and **input/capture/driver APIs** at the bottom. A strong automation tool uses the highest layer available for the task, and only falls downward when required. [22]

API / mechanism	Capabilities	Limitations	Permission / privilege notes	Typical use cases	Sources
UI Automation	Discover elements as a tree; query properties; invoke control patterns; subscribe to events	Coverage depends on provider quality; custom-drawn or poorly instrumented controls may be weak; threading must be handled carefully	User-level works for same-IL apps; elevated/ protected UI requires UIAccess/ admin conditions	Primary selector system for Win32, WinForms, WPF, UWP/Store-style apps where providers exist	23
IUIAutomation COM interfaces	Direct COM access to root element, conditions, element queries, events, tree walking	Lower-level and more verbose than wrappers; COM/threading complexity	Same boundary considerations as UIA generally	Native service implementation, high-performance querying, event subscriptions	24
MSAA / IAccessible	Legacy accessible object model with common attributes and behaviors	Smaller fixed property/action surface than UIA; weaker semantic/action model	Works broadly on older apps; still subject to Windows security boundaries	Legacy app coverage, compatibility fallback, inspection	25
IAccessible2	Extends MSAA with richer information for accessible objects	Not the native “one standard” Windows layer; depends on app/browser support	App-specific support; useful where IA2 is implemented	Firefox and some rich document/ browser tooling; bridging gaps left by UIA	26

API / mechanism	Capabilities	Limitations	Permission / privilege notes	Typical use cases	Sources
IAccessibleEx	Bridge from MSAA objects into UIA properties/ patterns without a full provider rewrite	Transitional rather than complete; only as good as provider implementation	Same accessibility trust boundary rules apply	Upgrading legacy controls, bridging mixed accessibility stacks	27
SetWinEventHook	Subscribe to accessibility/ window events across processes; scope by process/ thread; in- or out-of-context delivery	Requires a message loop; out-of-context delivery is queued; bitness and console-process cases affect delivery behavior	Cross-process eventing on the current desktop; not a general write/ control API	Event-driven waits, focus/ window-change notifications, stale-element detection	28

API / mechanism	Capabilities	Limitations	Permission / privilege notes	Typical use cases	Sources
HWND / Win32 window functions	Enumerate windows, move/resize/ reorder windows, manage z-order and visibility	Window-level only; little semantic control of internal widgets	Same desktop/ session assumptions; some focus rules still apply	Window discovery, attach-by-title/class, geometry management	29
SendMessage	Synchronous delivery to a window procedure; useful for app-specific commands and some control interactions	App/ message specific; caller blocks until processed; not a generic input replacement	Subject to UIPI; only same or lower integrity targets	Safe, explicit control when you know the message contract	30

API / mechanism	Capabilities	Limitations	Permission / privilege notes	Typical use cases	Sources
PostMessage	Asynchronous posting to a thread message queue	Async only; pointers forbidden for async functions below `WM_USER`; custom marshalling needed for messages `>= WM_USER`; queue limits exist	Subject to UIPI; access-denied on blocked posts	Low-coupling notifications, app-specific async messaging	31
SendInput	Inject keyboard/ mouse events serially into the input stream	Focus-sensitive; not target-specific on its own; does not reset keyboard state; blocked by UIPI with poor diagnostics	Can inject only into equal or lower integrity targets	Universal fallback for click/type/ drag/hotkeys	32
Raw Input	Receive low-level HID input data and device-specific events	Monitoring only, not injection; app must register devices to receive `WM_INPUT`	Standard user mode for registered devices	Input monitoring, advanced key/mouse observation, telemetry	33

API / mechanism	Capabilities	Limitations	Permission / privilege notes	Typical use cases	Sources
SetWindowsHookEx	Install thread/ global hooks; can inject DLLs into target processes depending on hook type	Bitness-sensitive; fragile; requires message pumping; risky for broad automation exposure	Cross-process hooks depend on process architecture and normal security boundaries	Monitoring, special integrations, niche interception —not a first choice for standard automation	34
DWM thumbnails	Live thumbnail relationship between source and destination windows	Visual only, not controllable semantics	Standard desktop API use	Visual previews, operator UIs, verification dashboards	35
Desktop Duplication	Fast frame-by-frame desktop capture via DXGI surface, GPU-friendly	Capture only; whole-output oriented; no inherent semantics	Standard desktop capture path	Real-time screen capture for OCR/CV or remote collaboration	36
Windows.Graphics.Capture	Secure capture of windows/ displays, with system-backed capture APIs	Modern Windows-only path; Win32 interop availability depends on Windows 10 version	Requires supported Windows 10+ environment	Per-window capture, snapshots, modern capture backend	37

API / mechanism	Capabilities	Limitations	Permission / privilege notes	Typical use cases	Sources
Virtual HID Framework / HID minidrivers	Present synthetic HID devices to Windows through driver infrastructure	Far higher complexity and operational risk than user-mode APIs; capture/ control semantics move into driver space	Driver-development/ deployment path; unsuitable as the starting point for a general OSS controller	Specialized research/test rigs, highly constrained anti-cheat or device emulation scenarios	[38]

Two security facts dominate the entire stack. First, **UIPI** blocks lower-integrity processes from sending many messages and injected input to higher-integrity targets; this shows up explicitly in `SendMessage`, `PostMessage`, and `SendInput`. Second, Microsoft's **UIAccess** path is deliberately narrow: a UIAccess app must be signed, installed in a secure location such as Program Files, and carry the correct manifest flag; even then, it is not a magic bypass for SYSTEM-level UI. This is why a production controller should never assume "desktop control" implies "full desktop control." [39]

From an engineering perspective, the decision tree is straightforward. Use **UIA patterns first** for button invocation, value setting, selection, toggles, and expansion. Use **Win32 window APIs** for host-window lifecycle and geometry. Use **message APIs** only when the target application's message contract is known and stable. Use **SendInput** as the general fallback when semantic control is unavailable. Use **capture APIs plus OCR/CV** when neither accessibility nor messaging is sufficient. Driver-level injection should remain explicitly out of scope until the far end of the roadmap. [40]

Screen understanding models and toolkits

For real-time Windows screen understanding, the right question is not "which single OCR engine is best?" but rather "which **mix** of text recognition, layout parsing, and GUI-object grounding **minimizes latency for the current step?" In practice, a desktop controller should prefer** ROI-first perception**: first use accessibility or window metadata to narrow candidates, then run OCR or icon detection only on those regions, and reserve full-screen parsing for inaccessible apps or ambiguous states. That is how modern agent work such as OmniParser becomes practical rather than prohibitively expensive. [41]

Toolkit / model family	Best fit	Strengths	Trade-offs	Integration notes for a Windows GUI tool	Sources
Tesseract 5	Low-dependency printed OCR on cropped regions	Mature, offline, Apache-2.0, broad language support, LSTM engine and official traineddata sets	Less convenient than modern deep OCR stacks for difficult scene/UI text; not a layout engine by itself	Good baseline fallback for ROI OCR, offline deployments, and environments where deep dependencies are undesirable	42
EasyOCR	Quick Python OCR for screenshots and scene text	Very easy setup; supports 80+ languages; works on scene text and dense text	Less structured than full document pipelines; heavier than classic OCR for minimal installs	Good for rapid prototyping and multilingual fallback OCR in Python-centric stacks	43
PaddleOCR / PP-OCRv5	Best default open-source OCR pipeline for product use	Strong multilingual support; explicit mobile/server model split; repo emphasizes structured JSON/Markdown output and document parsing	Broader dependency footprint than Tesseract; full stack can be more than needed for tiny ROI tasks	Best overall default for OCR service layer; use mobile models for ROI OCR and server models only when accuracy needs justify them	44
docTR	Python-first OCR with reading-order/document export focus	Good ergonomic API and end-to-end OCR pipeline; actively maintained	More document-oriented than GUI-specific; not purpose-built for icon grounding	Strong choice when screenshots include document-like panes, PDFs, or complex text blocks	45
LayoutParser	Layout region detection / document structure	Clean API for deep layout parsing; pairs well with OCR	Not an OCR engine by itself; best for structural segmentation	Useful when UI surfaces resemble forms, reports, PDFs, or dense dashboards	46

Toolkit / model family	Best fit	Strengths	Trade-offs	Integration notes for a Windows GUI tool	Sources
MMOCR	Custom OCR research/ deployment toolbox	Rich model zoo for text detection, recognition, and downstream tasks; OpenMMLab infra	Heavier engineering/ setup cost than turnkey OCR libraries	Best when you need to train or swap detectors/ recognizers rather than just consume OCR	47
CRAFT	Text detection for arbitrarily shaped scene text	Strong detector for irregular/ curved text; widely used in practical OCR stacks	Detector only; must be paired with a recognizer	Good region proposal model when UI text is stylized or laid out irregularly	48
DBNet / DBNet++	Lightweight real-time text detection	Strong speed/ accuracy trade-off; paper reports 62 FPS with a ResNet-18 backbone on MSRA-TD500	Detector only; needs recognition stage	Good custom detector if you need a fast CPU/ GPU text-proposal stage for screen OCR	49
CRNN	Compact recognizer for custom OCR stacks	Still one of the simplest effective recognizers for image-based sequence recognition	Older architecture than transformer recognizers	Excellent lightweight recognizer paired with DBNet/CRAFT in a custom screen-text OCR pipeline	50
TrOCR	Higher-accuracy transformer recognizer	Strong end-to-end transformer OCR; Microsoft-origin checkpoints are well supported in Transformers	Heavier than CRNN-style recognizers; less suitable for very tight latency budgets	Use when OCR quality matters more than minimal CPU/ GPU load, especially on difficult cropped text	51

Toolkit / model family	Best fit	Strengths	Trade-offs	Integration notes for a Windows GUI tool	Sources
OmniParser	GUI element/ icon detection and grounding	Specifically designed for UI screenshot parsing into structured elements; directly relevant to LLM action grounding	Not a deterministic control API; vision-only parsing is still probabilistic	Ideal as a fallback CV layer for icon-only controls, inaccessible apps, or screenshot-only agent modes	[52]

A practical recommendation for the proposed tool is a **three-tier perception stack**. Tier one should be **no OCR at all** when UIA/MSAA/IA2 already provide element text and structure. Tier two should be **fast ROI OCR**, with PaddleOCR mobile models as the mainline option and Tesseract as the lowest-dependency fallback. Tier three should be **full screenshot parsing**, using OmniParser or a custom detector when the app is inaccessible or the relevant affordance is icon-only. LayoutParser and docTR are best treated as optional structure-aware modules for document-like panes rather than as the default path for every click. [53]

The main performance lesson from the open-source OCR literature is that **detector and recognizer can be decoupled**. A lightweight detector such as DBNet with a compact recognizer such as CRNN is still a very credible low-latency stack. Transformer recognizers such as TrOCR can raise accuracy but should be used selectively on hard crops or post-failure retries rather than on every frame. That design keeps average-step latency low while still improving recovery on difficult screens. [54]

Primary references and recent literature

The most useful primary sources divide into four buckets: Microsoft platform documentation, protocol/ interface specifications, project docs for practical open-source libraries, and recent research on multimodal desktop agents. The list below emphasizes sources from the last five years when possible, while retaining a few earlier foundational OCR papers that are still directly relevant to lightweight implementation choices.

Source family	Why it matters	Sources
Microsoft accessibility and automation docs	Authoritative reference for UIA, MSAA, IAccessibleEx, control patterns, properties, eventing, and security boundaries	[55]
Microsoft input/windowing docs	Ground truth for SendInput, message APIs, hooks, window management, Raw Input, and DWM	[56]
Microsoft capture docs	Best source for capture backend selection on Windows 10+	[57]
MCP and WebDriver specifications	Best models for agent interoperability and language-neutral remote control semantics	[4]

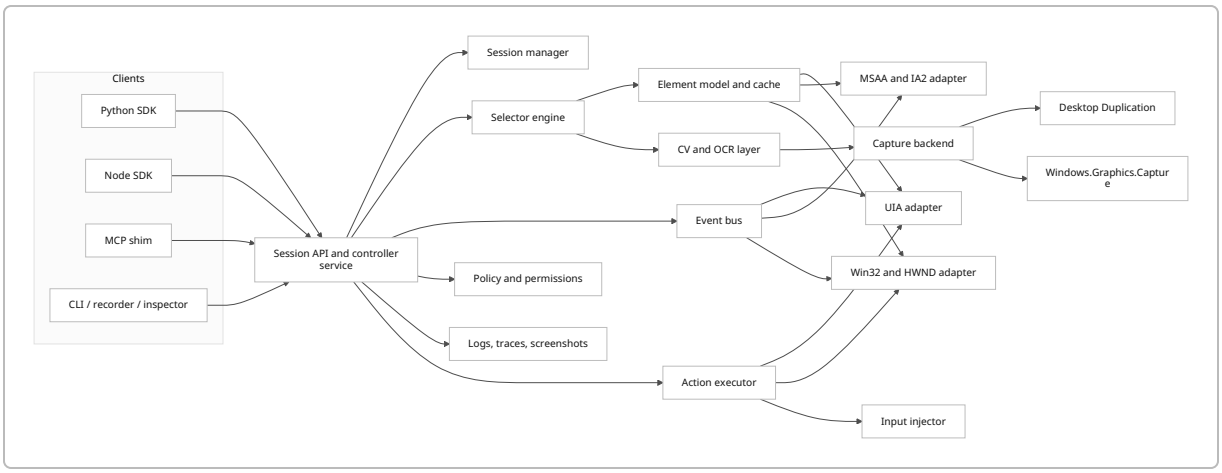
Source family	Why it matters	Sources
Recent agent papers	Show the present-day benchmark and architecture direction for real-computer agents	[58]
OmniParser	Most relevant recent screen-grounding paper/project for screenshot-first GUI agents	[52]
OCR/layout toolkits	Practical open-source building blocks for production perception stacks	[59]
OCR foundation papers	Still the key references for lightweight detection/recognition trade-offs	[60]

Three recent research threads are especially relevant to an MCP-like Windows GUI controller. **OSWorld** argues that real-computer benchmarking is necessary because standard web-only or narrow task benchmarks do not capture genuine computer-use complexity. **Windows Agent Arena** narrows that lens to the Windows ecosystem and underlines the need for realistic Windows application coverage. **OmniParser** shows that pure visual grounding materially improves screenshot-based action prediction, especially when semantic accessibility data is missing or incomplete. Together these sources strongly support a controller that exposes structured elements when possible, but always retains a screenshot/perception fallback. [61]

Project documentation reinforces the same conclusion from the implementation side. Microsoft's UI Automation docs make clear that the platform already offers an object model, properties, control patterns, and events; open-source libraries such as pywinauto and FlaUI mainly become successful by making those APIs pleasant to use. The newer agent/MCP projects become useful when they package the same primitives into tool-oriented transports and add screenshot/state handling, not when they replace the underlying semantics. [62]

Reference architecture and implementation roadmap

The right architectural goal is **not** "an MCP server that clicks on things." It is a **deterministic Windows controller service** with a stable element model, a selector engine, robust waiting/error semantics, and a narrow privileged surface. MCP should sit on top as one transport and orchestration layer, not as the only interface. That parallels how WebDriver separates the browser automation engine from the client bindings, while MCP defines how LLM applications discover and call external tools. [63]



The controller service should be local to the target Windows desktop and expose a language-neutral API over **JSON-RPC or HTTP/WebSocket**, with generated SDKs for Python and Node. The core implementation language should be **C#/.NET** unless there is a strong need for lower-level native code: Windows accessibility and COM interop are mature there, Win32 bindings are solid, and producing an OSS Windows-first service is faster than starting in Rust or C++. A small native helper can still be added later for edge capture or input cases, but it should not be the architectural center.

The **core components** should be these:

Component	Responsibility	Recommended approach
Session manager	Attach to apps/windows, manage lifecycle, isolate clients	One process-local controller service, multiple sessions by window/process
Element model	Stable IDs, bounds, ancestry, source backend, supported actions	Cache UIA/MSAA/IA2/Win32 facts plus confidence and staleness metadata
Accessibility adapter	Primary query path and action semantics	UIA first; MSAA/IAccessible2 bridge where UIA is weak
Win32 adapter	Window enumeration, focus, resize, z-order, app/process attach	Wrap HWND APIs and selected message-based operations
Capture layer	Window/display frames for OCR and debugging	Prefer Windows.Graphics.Capture when available; DXGI Desktop Duplication fallback on Windows 10+ [64]
CV/OCR layer	OCR, icon grounding, layout parsing	PaddleOCR mobile + Tesseract fallback; OmniParser optional GUI grounding layer [65]
Input injector	Click, type, drag, hotkeys	Prefer UIA patterns; fall back to SendInput; use messages only for known contracts [66]

Component	Responsibility	Recommended approach
Event bus	Window and element change notifications	SetWinEventHook + UIA events where available [67]
Security / policy	Capability allowlists, redaction, elevation control	Default non-elevated, separate helper for privileged scenarios, full audit trail
Logging / telemetry	Reproducibility and failure analysis	Structured action logs, selector traces, screenshots, timings, element snapshots

The **selector strategy** should be hybrid and explicit. The selector language should support accessibility-first queries such as `role=Button[name="OK"]`, `automationId=saveBtn`, and `path=/Window/Edit[1]`, plus spatial/contextual selectors such as `below(text("Username"))`, and finally vision selectors such as `ocr:"Save As"` or `icon:"trash"`. Resolution order should be:

1. **Accessibility-first** using UIA properties and control patterns.
2. **Legacy bridge** using MSAA/IAccessible2 when UIA is incomplete.
3. **Win32 context refinement** using host window, class name, or HWND ancestry.
4. **Vision fallback** using OCR/icon detection on the active window or a narrowed ROI.
5. **Coordinate-only fallback** only when all semantic strategies fail.

That order is justified by Microsoft's richer UIA property/pattern model and by the fact that screenshot-only approaches are inherently probabilistic. [68]

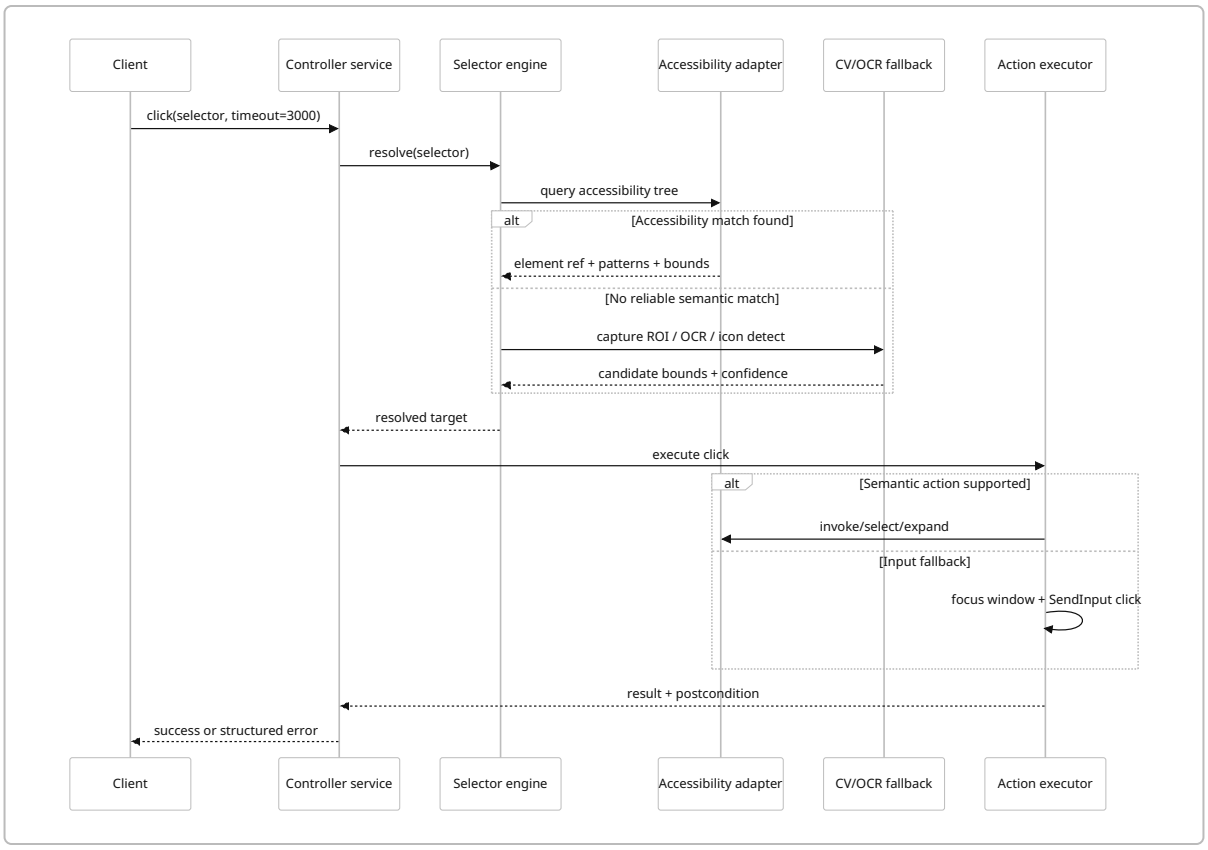
A minimal but credible **API surface** would include the following methods:

Method / endpoint	Behavior	Notes
`createSession()`	Start or attach a controller session	Supports attach by pid, title, class, executable
`listWindows()`	Enumerate top-level windows with metadata	Uses Win32 window APIs
`snapshot()`	Return current structured state	Tree + visible windows + optional screenshot
`query(selector)`	Resolve one or more elements	Returns stable element refs and confidence
`getElement(ref)`	Read current properties/ patterns/bounds	Revalidates staleness
`click(ref or selector)`	Perform click or invoke	Prefers semantic pattern actions before pointer injection
`type(ref, text)`	Set or enter text	Prefer ValuePattern/set semantics, else focus + SendInput

Method / endpoint	Behavior	Notes
`press(keys)`	Global or window-scoped shortcuts	Uses SendInput/hotkey sequencing
`setWindowState()`	Focus, move, resize, minimize, restore	Win32 + SetWindowPos layer
`scroll()` / `drag()`	Pointer-driven interactions	Input injector with optional semantic scroll pattern
`screenshot()`	Window/display/element capture	Capture backend abstraction
`waitFor()`	Wait for selector/event/state	Event-driven when possible, polling fallback
`subscribeEvents()`	Async stream of focus/ window/tree changes	WebSocket or server-sent stream
`getLogs()` / `getTrace()`	Read structured action traces	First-class debugging support

The service should expose **sync and async** behavior clearly. All action methods should be synchronous by default from the caller's perspective, returning only after the action has been dispatched and optional wait conditions have been satisfied or timed out. Event streams should be asynchronous over WebSocket. That yields a programming model closer to Playwright than to ad hoc scripting.

A representative **sequence flow** should look like this:



The **error model** should be structured, not textual. At minimum, define stable error codes such as `ElementNotFound`, `AmbiguousMatch`, `StaleElement`, `ActionNotSupported`, `FocusDenied`, `PrivilegeBoundary`, `WindowClosed`, `VisibilityMismatch`, `LowConfidenceVisionMatch`, and `TimeoutExceeded`. Each error should include the original selector, the backend attempted, candidate matches, screenshot/trace IDs, and recovery hints. This is especially important for LLM agents, which need machine-readable failure reasons.

A good initial **performance budget** is best expressed as proposed targets, not as guarantees:

Metric	Proposed target
Window enumeration	p95 under 30 ms on a local desktop
Accessibility query for a modest tree	p95 under 100 ms
Screenshot of a single window	p95 under 40 ms
ROI OCR on a small crop	p95 under 120 ms with lightweight models
End-to-end click dispatch	p95 under 250 ms excluding app response
Full `snapshot()` with tree and screenshot	p95 under 200 ms on ordinary business apps

The **security model** should be conservative. The daemon should run non-elevated by default, expose only explicit capabilities, and require per-session approval or policy to access file system, PowerShell, clipboard,

or remote transports. If privileged automation is needed, ship a **separate helper** rather than running the whole service elevated. Any future UIAccess mode should follow Microsoft's documented requirements—signed binary, secure install location, manifest flag—and should be framed as an accessibility-compatible advanced feature, not a default operating mode. [69]

A small **integration example** for Python could look like this:

```python
from wingui import WindowsController

ctl = WindowsController.connect()
session = ctl.attach(window_title="Untitled - Notepad")

editor = session.query('role=Edit')
editor.type_text("hello from Python")

session.query('role=MenuItem[name="File"]').click()
session.query('hybrid(role=MenuItem,name="Save As")').click()
```

And for Node:

```javascript
import { connect } from "@wingui/client";

const ctl = await connect();
const session = await ctl.attach({ title: /Notepad/ });

await session.query('role=Edit').typeText("hello from Node");
await session.query('role=Button[name="OK"]').click();
```

For an **MCP shim**, the top-level tools should be coarse enough for agent planning but still deterministic: `snapshot`, `query`, `click`, `type`, `press`, `wait_for`, `move_window`, `screenshot`, and `read_logs`. Do not expose raw unrestricted desktop actions without a policy layer. The MCP server should be a broker over the controller service, not the only implementation surface.

The **prioritized roadmap** should be:

Priority	Milestone	Deliverables	Why this order
Highest	Accessibility-first local controller	Session manager, UIA queries, Win32 window attach, click/type/wait, Python SDK	Delivers the most reliability with the least ambiguity
High	Traceability and robustness	Inspector UI, structured logs, event hooks, stale-element recovery, Node SDK	Makes the tool debuggable and usable in CI/agent loops

Priority	Milestone	Deliverables	Why this order
High	Capture and perception fallback	DXGI/WGC capture, ROI OCR, hybrid selectors, screenshot traces	Expands support to inaccessible or partially accessible apps
Medium	MCP shim and remote mode	MCP wrapper, WebSocket/SSE transport, auth, session-scoped policies	Enables agent ecosystems without destabilizing the core
Medium	Legacy/browser fallbacks	MSAA/IA2 bridge, browser-specific adapters where needed	Covers Firefox/web/document edge cases that UIA alone misses
Lower	Recorder and benchmark integration	Recorder/replayer, replay diffing, Windows Agent Arena / OSWorld adapters	Useful once the core semantics are stable
Lowest	Driver-level experimentation	Optional research path with virtual HID or custom drivers	High operational cost and security risk; defer until justified

The **risk analysis** is manageable but real:

Risk	Why it matters	Mitigation
Incomplete UIA coverage	Custom controls and legacy apps may expose weak trees	Keep MSAA/IA2 bridge and vision fallback
UIPI / elevation boundaries	Actions may silently fail against elevated windows	Make privilege state explicit; separate helper; structured `PrivilegeBoundary` errors
Focus and race conditions	The wrong window may receive `SendInput`	Re-check foreground window before input; prefer semantic patterns
DPI, scaling, theming, localization	Vision selectors and coordinates drift easily	Normalize coordinate spaces; use semantic selectors first; keep OCR/icon fallback language-aware
Browser edge cases	Firefox and some embedded content expose different accessibility paths	Maintain IA2 fallback and per-app adapters where necessary
Security exposure	A remote desktop controller can become an abuse surface	Capability allowlists, auth, sandboxed remote transport, full audit logging
Maintenance sprawl	Supporting every niche API raises complexity quickly	Keep the core narrow and plugin-based; defer exotic paths

Risk	Why it matters	Mitigation
Driver-level ambitions	Kernel- or driver-level work sharply raises install and maintenance burden	Keep out of MVP and major releases unless a concrete user need appears

Open questions and limitations remain. I did not fully verify the exact SPDX license or latest commit date for every niche project surfaced in the ecosystem, especially some RPA platforms and continuation/fork projects; where that was not high-confidence, I marked it explicitly rather than guessing. Performance numbers for OCR and desktop capture vary heavily by hardware, DPI, app framework, and whether capture is whole-screen or window-scoped. And some wrapper libraries do not clearly document their exact underlying API in the top-level project page, even when their behavior strongly implies UIA-backed operation.

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